CardioTwin 3D: Interpretable Vessel-Level Coronary Risk Mapping

Multimodal AI Hackathon 2026 - Track A: Cardiovascular Risk Visualization & Prediction. Team Akatsuki. Research prototype for education and decision support only; not a diagnosis and not a substitute for angiography or clinician judgement.

Abstract

CardioTwin 3D predicts overall coronary artery disease (CAD) and stenosis of the LAD, LCX and RCA from 55 routine clinical, ECG, laboratory and echocardiographic measurements, paints each probability onto segmented BodyParts3D coronary meshes in an interactive 3D viewer, and explains every prediction with exact SHAP values. Models are validated on a locked 61-patient hold-out (seed 7) plus nested 5-fold cross-validation with bootstrap intervals. Locked ROC-AUC: CAD 0.926 [0.854-0.982], LAD 0.923 [0.849-0.979], LCX 0.806 [0.678-0.919], RCA 0.704 [0.561-0.827]. The system runs as one URL (FastAPI serves the built app), works without a GPU, and was developed on 303 referred patients with no external validation.

1. Problem & approach

Pre-test risk scores give one number for the whole heart. Clinicians reading echo, ECG and lipids want to know which territory is at stake and which findings drive the estimate. CardioTwin 3D gives both: four calibrated probabilities (CAD + 3 vessels), each painted on its artery (green <0.40, orange <0.70, red above), each explained measurement-by-measurement. Key design rules: (1) leakage features (LAD/LCX/RCA/Cath/CAD) are never inputs - asserted in code and unit-tested; (2) probabilities are Platt-calibrated with per-target Youden thresholds; (3) every number in the UI comes from live model inference - no hardcoded predictions; (4) limits are stated in-app, not hidden.

2. Data

UCI Extension of Z-Alizadeh Sani dataset (303 patients, Tehran referral centre, 71% CAD prevalence). 55 input features in 6 clinical groups: demographics (5), history & risk factors (12), examination (14), ECG (7), laboratory (14), echocardiography (3). Labels: vessel stenosis (Normal/Stenotic, >=50% narrowing); overall CAD derived as positive if any of LAD/LCX/RCA stenotic. No missing cells. The 55-field form exposes every feature with dataset-observed options, so each SHAP bar is editable. Data file SHA pinned via committed raw sheet; train/test split seed fixed at 7.

3. Methods

Candidates per target: L2/L1 logistic regression, random forest, gradient boosting (XGBoost), all SHAP-compatible, preprocessing inside pipelines (median/mode imputation, scaling, one-hot). Selection: inner 5-fold ROC-AUC on the 242-patient development set -> CAD: logistic L2, LAD: random forest (300 trees), LCX/RCA: XGBoost (300 trees, depth 4). Calibration: Platt (sigmoid), cross-fitted for evaluation, refit on all development data. Operating points: per-target Youden thresholds on out-of-fold predictions (CAD 0.65, LAD 0.60, LCX 0.40, RCA 0.325). Explanations: TreeExplainer for tree models, LinearExplainer otherwise, one-hot columns summed back to measurements, shown in calibrated-probability points with % shares. Frontend: React + React Three Fiber + FastAPI; segmented anatomy coloured per-vessel on one shared ramp; SHAP served by /explain.

Reproducibility: seeds fixed (42/7), requirements pinned, models + metrics + per-patient locked predictions committed (models/*.pkl, metrics_final.json, locked_test_predictions.csv). Leakage tests: tests/test_leakage.py (2 passed). Retrain: backend/train_locked.py (~2 min), backend/train_nested.py, backend/make_figures.py.

4. Results - locked 61-patient test (seed 7)

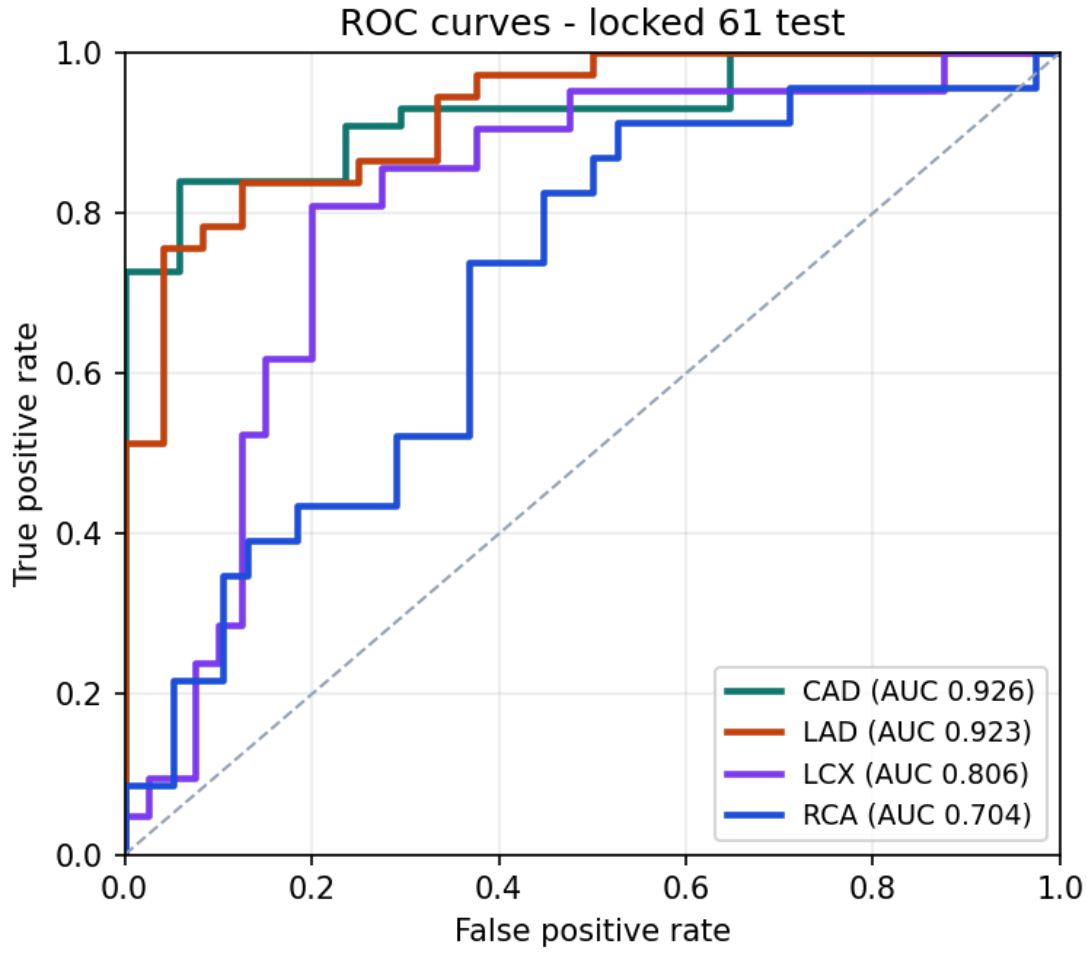
The table below is the number to trust: 61 stratified hold-out patients never seen in development, scored once, 2000-sample bootstrap CIs, Brier scores included.

	CAD	LAD	LCX	RCA
Model	LogisticRegression	RandomForestClassifier	XGBClassifier	XGBClassifier
ROC-AUC	0.926	0.923	0.806	0.704
Accuracy	0.836	0.836	0.754	0.639
Precision	0.925	0.886	0.607	0.514
Recall	0.841	0.838	0.81	0.783
F1	0.881	0.861	0.694	0.621
Brier	0.103	0.118	0.176	0.212
Threshold	0.65	0.6	0.4	0.325

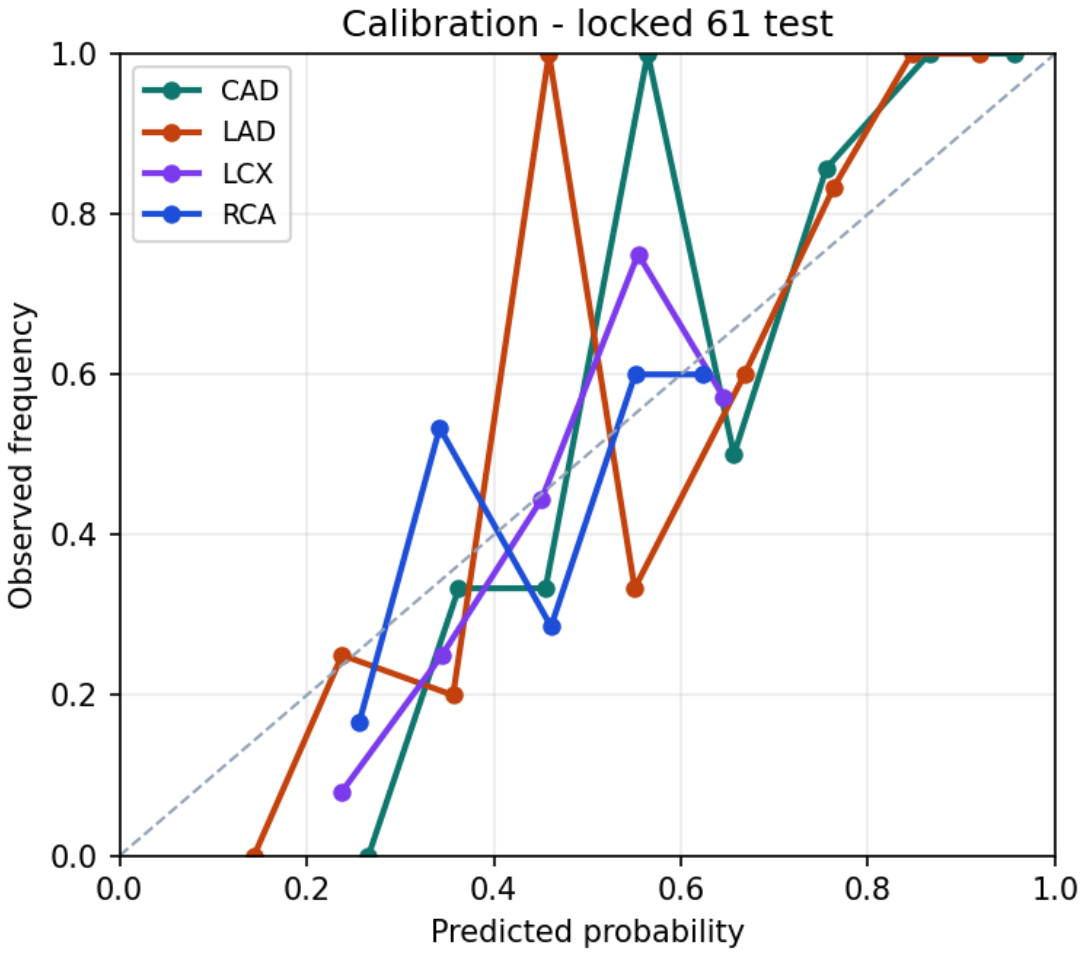
Nested 5-fold OOF reference (all 303): CAD AUC 0.927, LAD 0.849, LCX 0.732, RCA 0.712. CAD/LAD are informative; LCX/RCA are moderate - a data limit (many model classes plateau near AUC 0.74), not a modelling failure. Thresholds trade precision for recall on the weaker vessels; the UI shows each cut-off beside its probability.

4b. ROC & calibration (locked 61)

Curves computed from models/locked_test_predictions.csv (40-point ROC, 10-bin calibration).



ROC curves per vessel.



Calibration: observed vs predicted (diagonal = ideal).

5. System

3D visualization

BodyParts3D thorax/heart glTF with named coronary segments mapped to model targets (LAD: Coronary_LAD(+Septal); LCX: Coronary_LCX; RCA: 5 segments). Rotate/zoom, click vessel/pill/card (synced), Front/Left/Top presets, isolate mode, vessel-focus B&W (grey anatomy, neon coronaries), heartbeat animation (optionally PR-driven), layer toggles (lungs, ribs, skin, muscle), fullscreen overlay, printable report. Condition painting (clearly labelled illustrative, not imaging): smoker lung tint, hypertensive enlargement, LDL glow, low-EF weak beat.

Dashboard & integration

One URL serves UI+API. 303 patients searchable with angiogram truth beside live predictions; 55-field grouped form with validation; vessel dossiers (prob vs threshold, pushing-up/pulling-down top-3, exact SHAP waterfall Top12/all); Performance tab (locked + nested tables, live ROC/calibration); How-to-use first-run guide; dark theme. Median predict+explain under a minute on CPU; 3D renders on demand without a dedicated GPU.

6. Limitations & safety

1. Single-centre referred cohort (n=303, 71% CAD); no external validation - recalibration required elsewhere. 2. LCX/RCA weakly predictable from routine data (AUC ~0.71-0.81, wide CIs). 3. Whole-vessel probabilities; no lesion localization (dataset has none). 4. Generic adult anatomy, not patient-specific; condition painting is illustrative. 5. Calibration slopes on the locked split show some over-confidence. 6. What-if edits show model behaviour, not causal effects. The UI carries a permanent educational-use disclaimer; this system is not a medical device.

Attribution & reproducibility

Data: Extension of Z-Alizadeh Sani Dataset, UCI ML Repository (CC BY 4.0). Anatomy: BodyParts3D, Database Center for Life Science (CC BY-SA 2.1 JP); derived meshes redistributed under the same licence. Code original, MIT. (C) Akatsuki. Reproduce: pip install -r requirements.txt; python backend/train_locked.py; pytest tests -q; uvicorn backend.main:app --port 8000; cd frontend && npm install && npm run dev.